Social Media Sentiment Analysis for Product Feedback

Analyzing Textual Data for Sentiment Classification

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What is Sentiment Analysis?

Determining the emotional tone behind words to understand attitudes, opinions, and emotions.

Why is it important?

- Assists businesses in understanding customer opinions.
- Enhances market research and product analysis.
- Supports informed decision-making.

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Methodology

Scope:

- Utilize machine learning techniques for text classification and implement preprocessing steps to clean and prepare data.
- Evaluate model performance using appropriate metrics.

Data Collection:

- Sourced from Kaggle dataset - https://www.kaggle.com/datasets/kashishparmar02/social-media-sentiments-analysis-dataset/code

Data Preprocessing:

- Tokenization, stop-word removal, stemming/lemmatization.

Model Development:

- Implemented Random Forest Classifier, Support Vector Classifier, Logistic Regression, Multinomial Naive Bayes

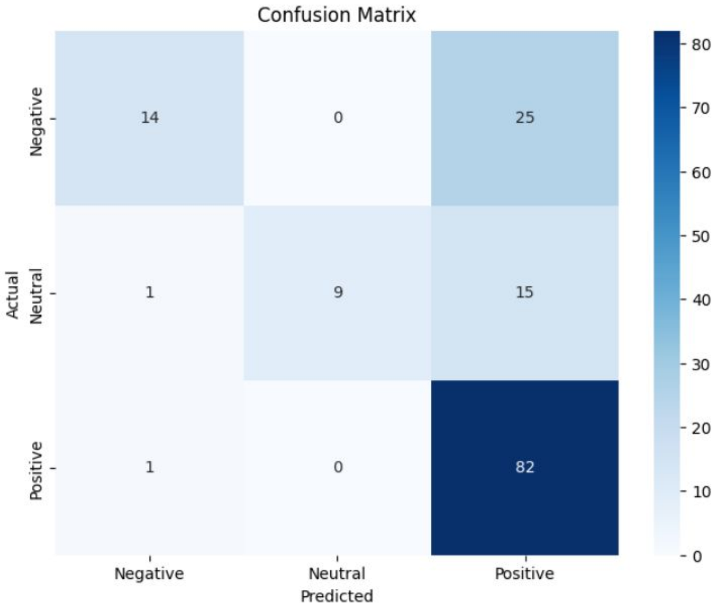
Evaluation Metrics:

- Accuracy, Precision, Recall, F1-Score.

Model Fitting

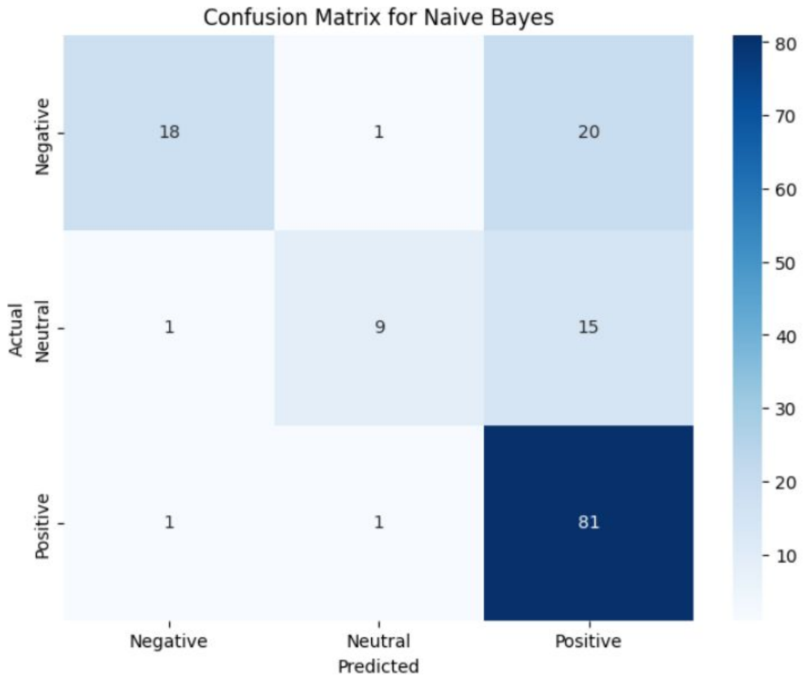
1. Logistic Regression

	precision	recall	f1-score	support
0	0.88	0.36	0.51	39
1	1.00	0.36	0.53	25
2	0.67	0.99	0.80	83
accuracy			0.71	147
macro avg	0.85	0.57	0.61	147
weighted avg	0.78	0.71	0.68	147



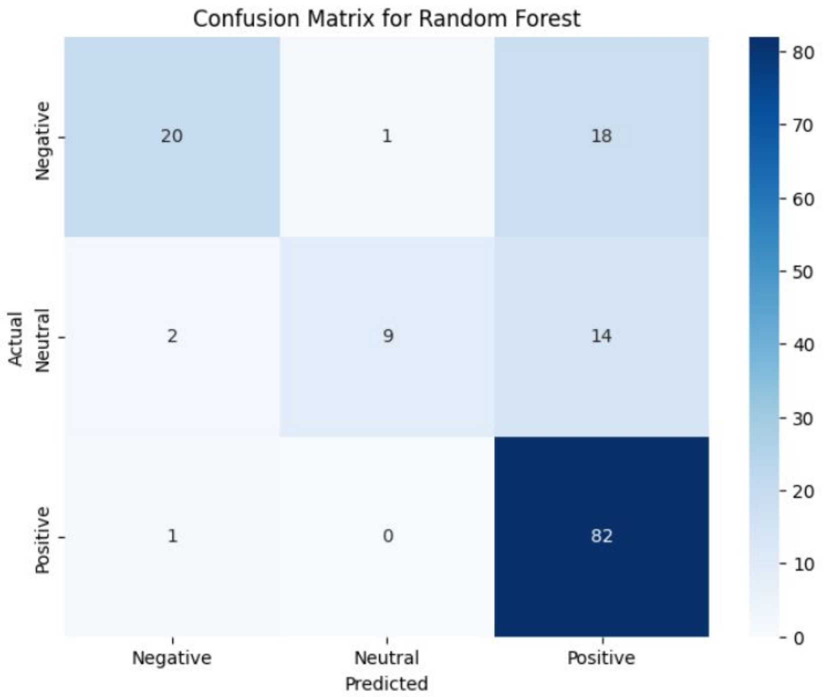
2. Naive Bayes

	precision	recall	f1-score	support
0	0.90	0.46	0.61	39
1	0.82	0.36	0.50	25
2	0.70	0.98	0.81	83
accuracy			0.73	147
macro avg	0.81	0.60	0.64	147
weighted avg	0.77	0.73	0.71	147



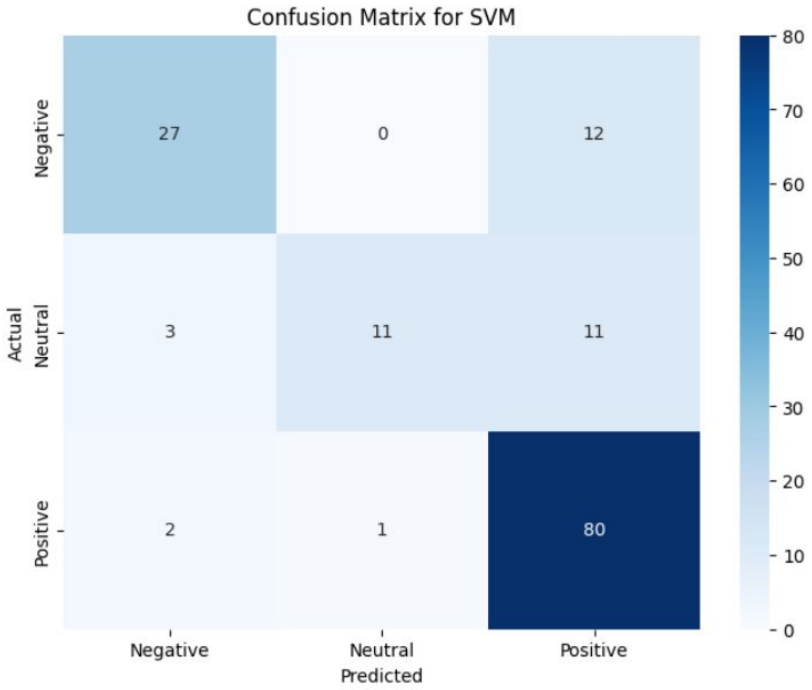
3. Random Forest

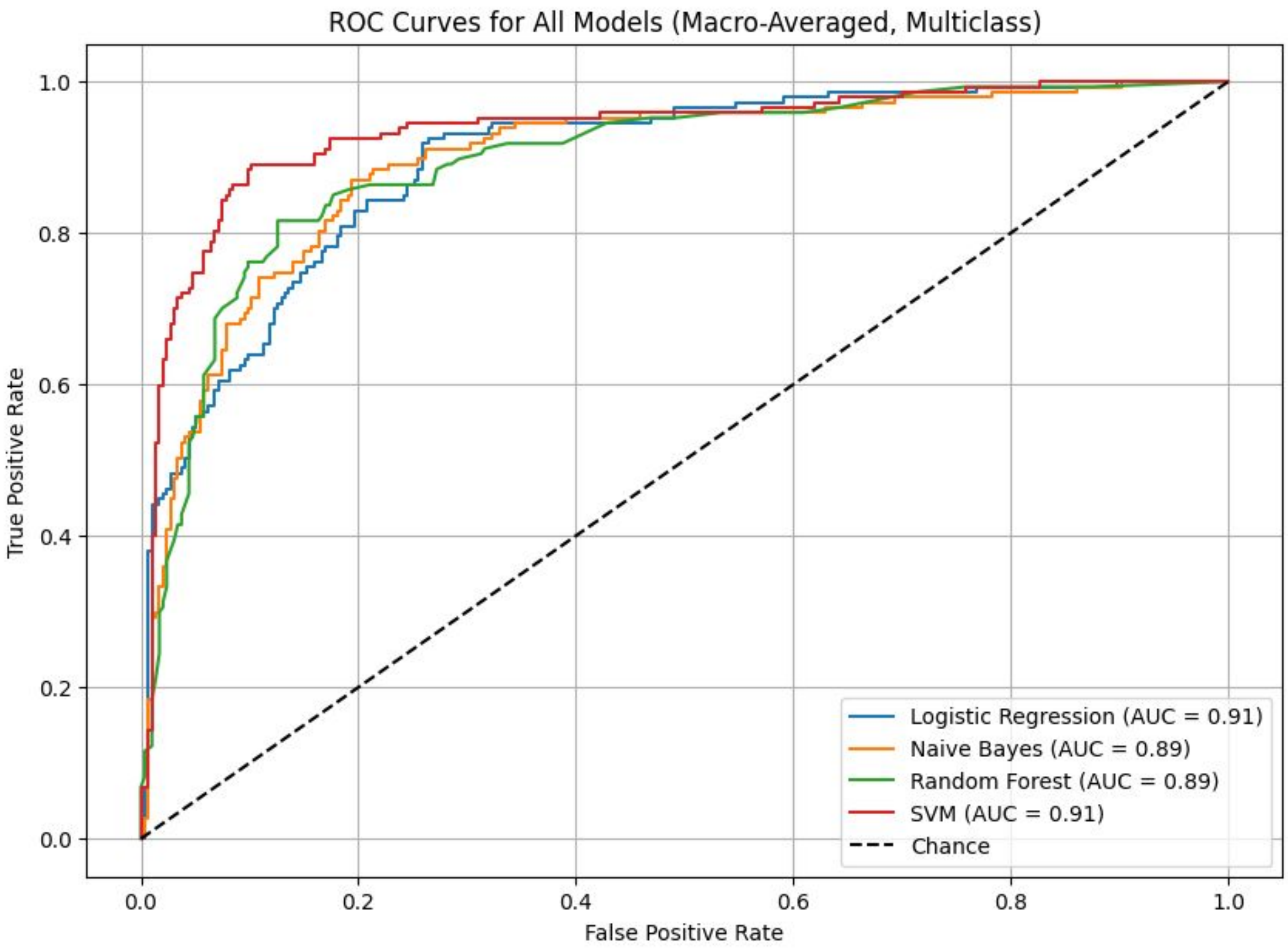
	precision	recall	f1-score	support
0	0.87	0.51	0.65	39
1	0.90	0.36	0.51	25
2	0.72	0.99	0.83	83
accuracy			0.76	147
macro avg	0.83	0.62	0.66	147
weighted avg	0.79	0.76	0.73	147



4. Support Vector Machine

	precision	recall	f1-score	support
0	0.84	0.69	0.76	39
1	0.92	0.44	0.59	25
2	0.78	0.96	0.86	83
accuracy			0.80	147
macro avg	0.85	0.70	0.74	147
weighted avg	0.82	0.80	0.79	147





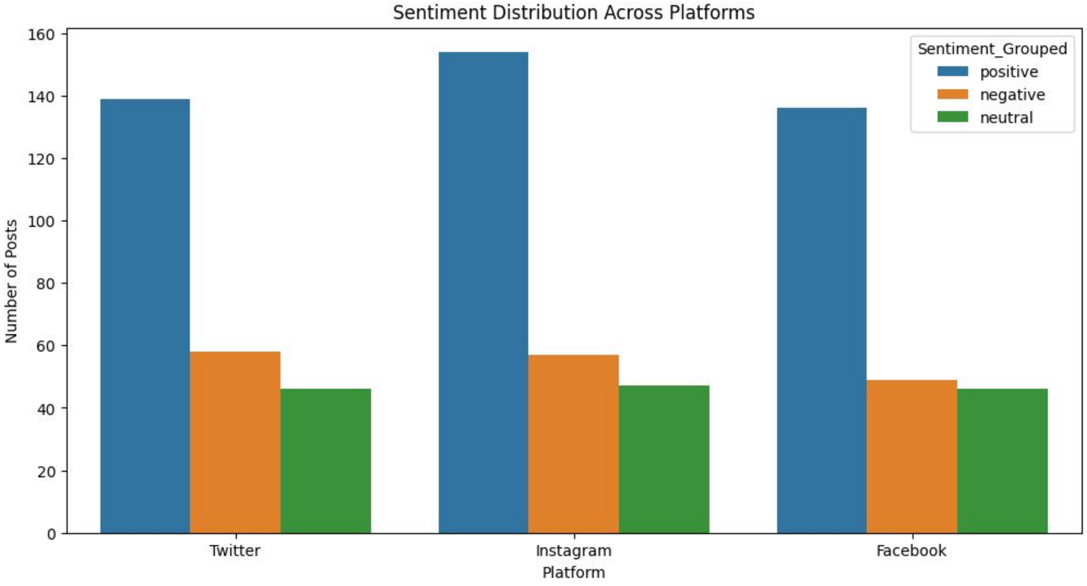
Model Accuracy Comparison:
Logistic Regression: 0.7142857142857143
Naive Bayes: 0.7346938775510204
Random Forest: 0.7687074829931972
SVM: 0.8027210884353742

```
Final Model Accuracy:
0.8027210884353742
              precision    recall
f1-score   support

           0       0.84      0.69
0.76        39
           1       0.92      0.44
0.59        25
           2       0.78      0.96
0.86        83

    accuracy
0.80       147
   macro avg       0.85      0.70
0.74       147
weighted avg       0.82      0.80
0.79       147
```

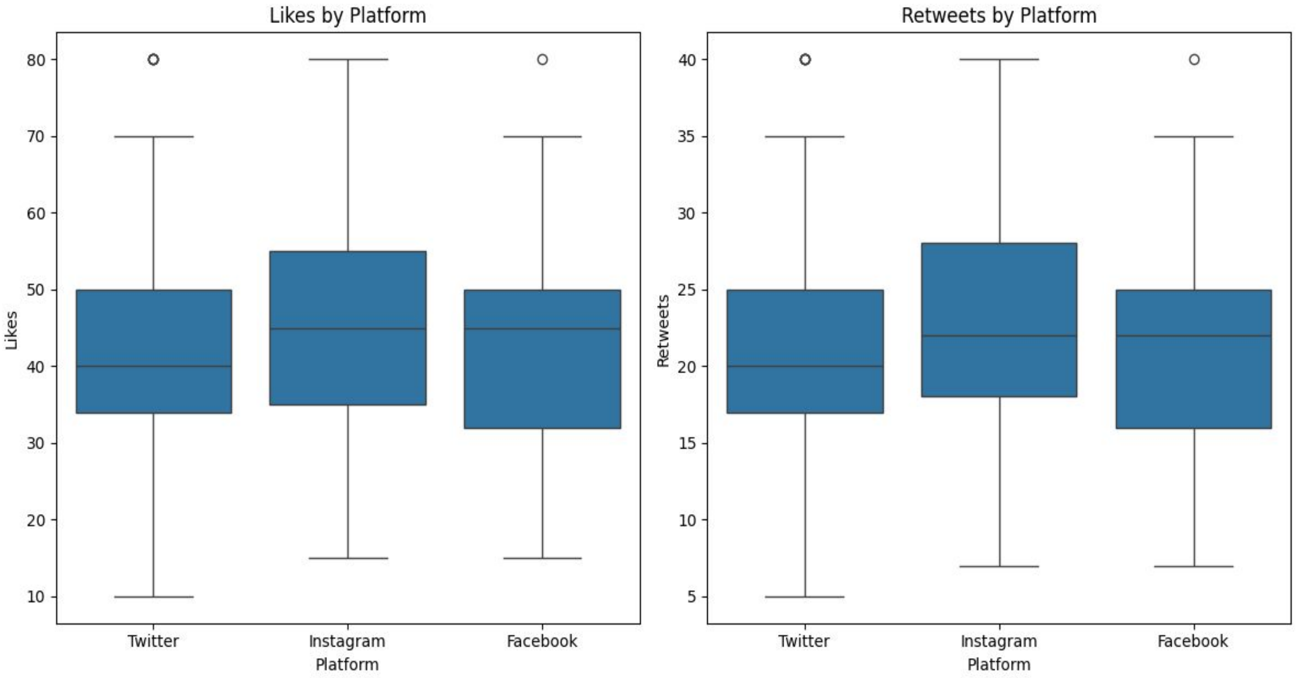
Hypothesis Testing



ANOVA F-statistic: 0.02111944604731679

ANOVA P-value: 0.9791744555731762

Fail to reject the null hypothesis. Sentiment distribution does not vary significantly across platforms.




```
-test results for Likes: Facebook vs Instagram
-statistic: -2.47173113257322
-value: 0.013786905956680346

-test results for Retweets: Facebook vs Instagram
-statistic: -2.497552165754012
-value: 0.012834338215203763
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-test results for Likes: Facebook vs Twitter
-statistic: 0.25950128521801297
-value: 0.7953616358342335

-test results for Retweets: Facebook vs Twitter
-statistic: 0.18273917312057006
-value: 0.8550811407695639
-------------------
-test results for Likes: Instagram vs Twitter
-statistic: 2.7123751548727704
-value: 0.006910955227809179

-test results for Retweets: Instagram vs Twitter
-statistic: 2.6711812418382768
-value: 0.007805390862656868
```

Hypothesis-**Is there a significant difference in likes/retweets across platforms?**

- Null Hypothesis: Mean engagement is the same on all platforms.
- Alternate Hypothesis: At least one platform differs in terms of mean user engegement.

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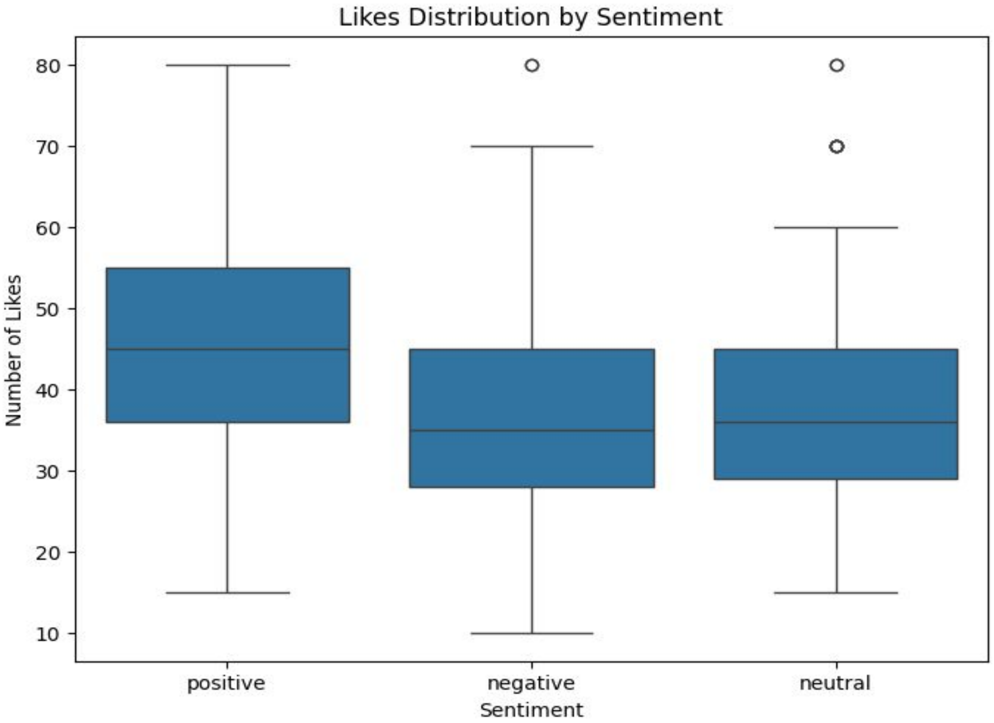


```
Reject the null hypothesis. At least one
sentiment group has a different average
number of likes.

Hypothesis 2: Likes differ by Sentiment
ANOVA p-value: 1.2213705078053843e-17
```

Hypothesis-**Do positive posts get more likes**

- Null Hypothesis: Likes differ significantly by sentiment.Positive posts get more likes than others.
- Alternate Hypothesis: Likes do not differ significantly by sentiment.

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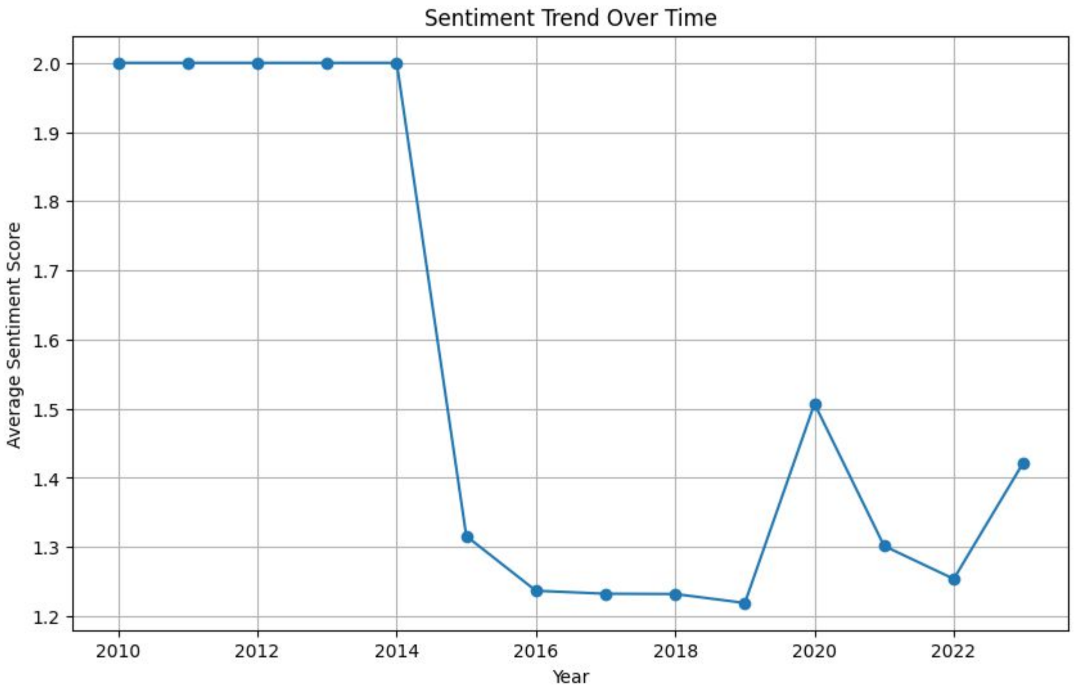
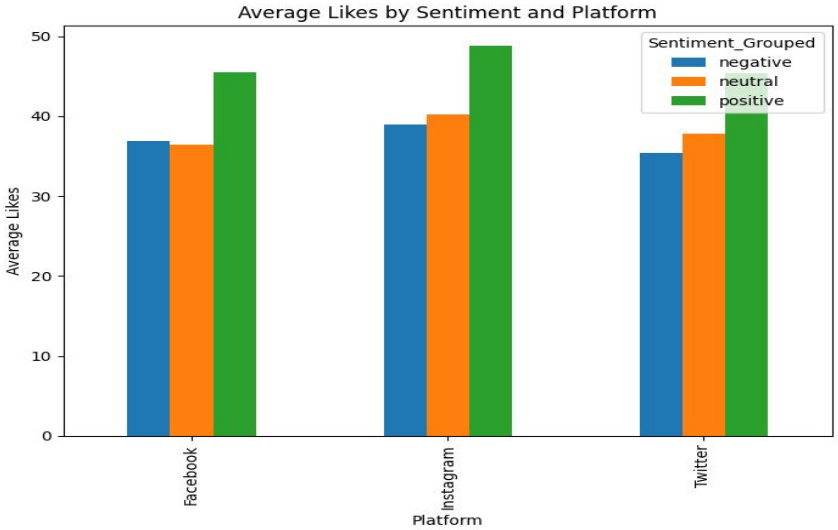
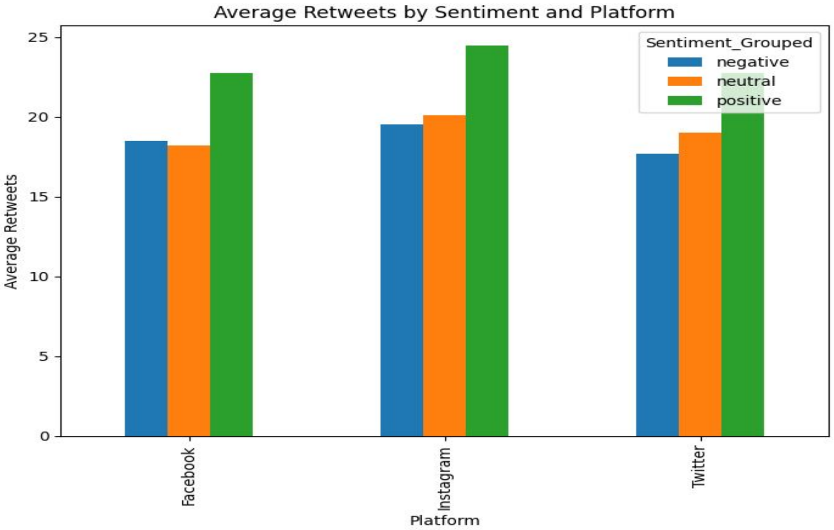


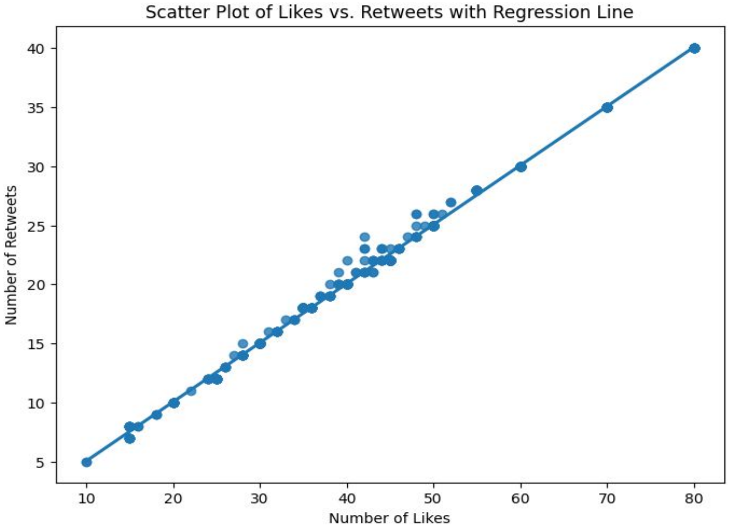

```
Reject the null hypothesis.
At least one year has a
different average sentiment
score.
ANOVA p-value:
1.2213705078053843e-17
              sum_sq      df
F    PR(>F)
C(Year)    14.989815   13.0
1.717368  0.05303
Residual  482.074392  718.0
NaN       NaN
```

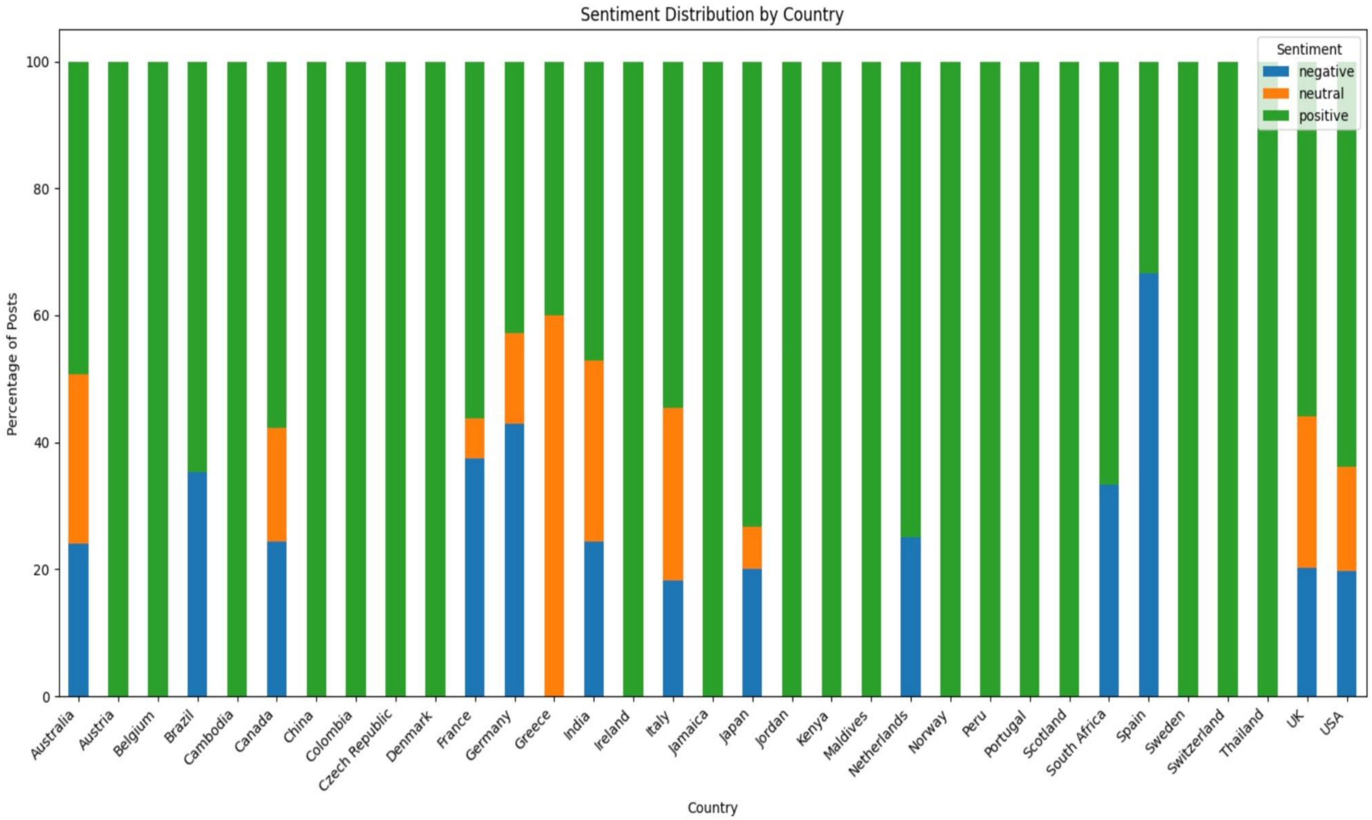
Hypothesis-**Does sentiment trend change over time**

- Null Hypothesis:The average sentiment scores are the same across all years.
- Alternate Hypothesis: At least one year has a different average sentiment score.


Average Likes by Sentiment and Platform
Sentiment_Grouped
negative
neutral
positive
Average Likes
Platform
Facebook
Instagram
Twitter


Average Retweets by Sentiment and Platform
Sentiment_Grouped
negative
neutral
positive
Average Retweets
Platform
Facebook
Instagram
Twitter


Scatter Plot of Likes vs. Retweets with Regression Line
Number of Retweets
Number of Likes


Sentiment Distribution by Country
Sentiment
negative
neutral
positive
Percentage of Posts
0
20
40
60
80
100
Australia
Austria
Belgium
Brazil
Cambodia
Canada
China
Colombia
Czech Republic
Denmark
France
Germany
Greece
India
Ireland
Italy
Jamaica
Japan
Jordan
Kenya
Maldives
Netherlands
Norway
Peru
Portugal
Scotland
South Africa
Spain
Sweden
Switzerland
Thailand
UK
USA
Country

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Challenges and Solutions

Challenges:

- Handling sarcasm and irony in text.
- Imbalanced dataset leading to biased predictions.

Solutions:

- Incorporated advanced NLP techniques to detect sarcasm.
- Applied resampling methods to balance the dataset.


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